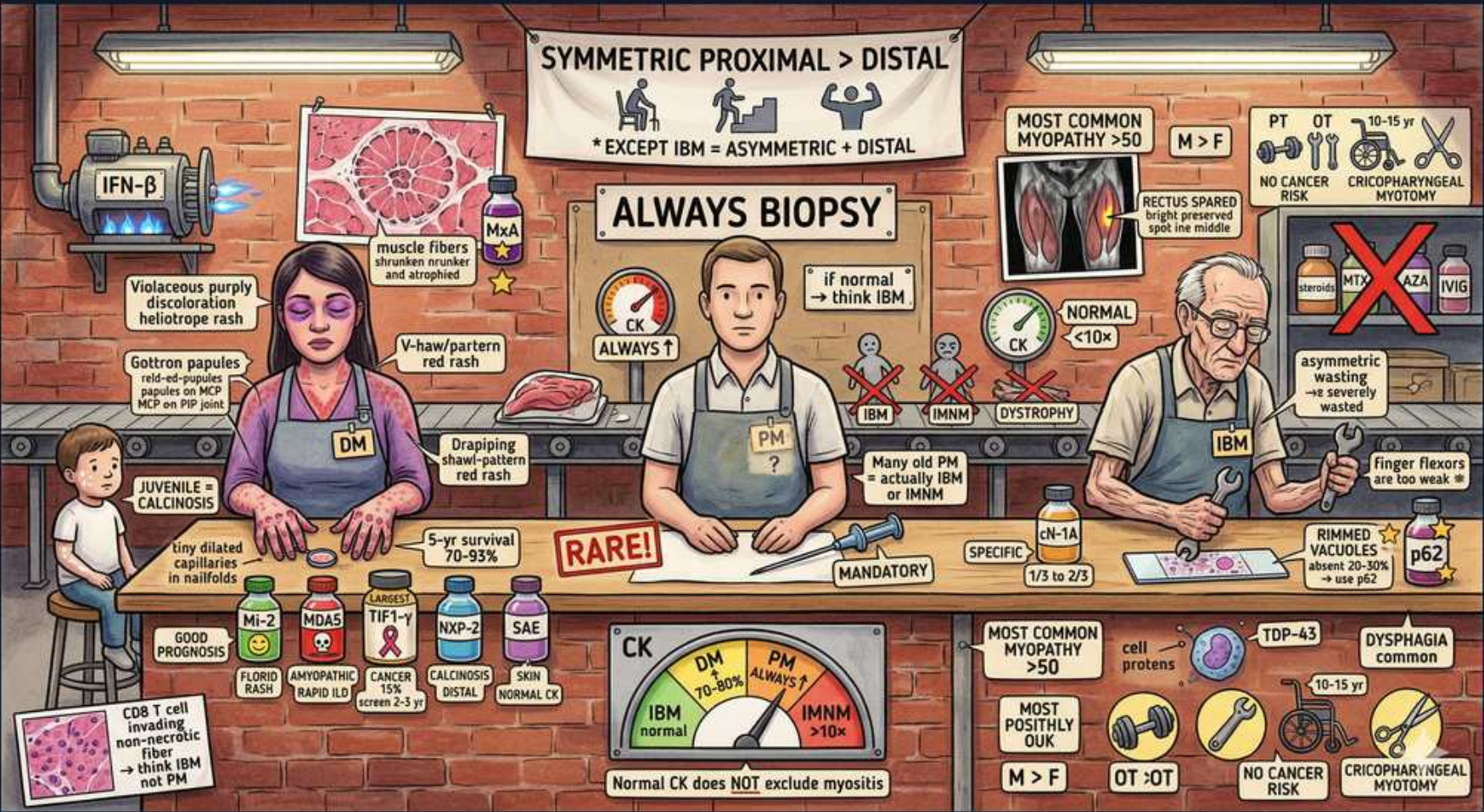


Inflammatory Myopathy — Dermatomyositis vs Polymyositis



1. Symmetric proximal > distal

WHAT YOU SEE

Top banner reading 'SYMMETRIC PROXIMAL > DISTAL *EXCEPT IBM: ASYMMETRIC + DISTAL' — the banner sets the defining motor pattern of DM/PM and flags IBM as the lone reversed-pattern exception.

WHAT IT ENCODES

The defining motor pattern of dermatomyositis and polymyositis is SYMMETRIC, PROXIMAL weakness that develops subacutely over weeks to months — trouble rising from a chair, climbing stairs, combing hair, lifting overhead. Bulbar/pharyngeal muscles can be involved (dysphagia, aspiration risk), but ocular muscles are spared (unlike myasthenia gravis). Sensation, reflexes, and the distal hands/feet are characteristically preserved early. Mnemonic: 'proximal, painless-ish, progressive, symmetric.'

BOARD TRAP

WRONG answer: choosing distal and/or ASYMMETRIC weakness (e.g., finger flexors and quadriceps in an older man) and calling it polymyositis. Discriminator: that pattern is inclusion body myositis (IBM), which is distal, asymmetric, slowly progressive, and steroid-refractory — the opposite of the symmetric proximal DM/PM pattern.

2. Heliotrope rash

WHAT YOU SEE

The woman on the left drawn with violaceous purple discoloration ringing both eyes — that periorbital lilac-purple shading IS the heliotrope rash, named for the purple flower.

WHAT IT ENCODES

Heliotrope rash = a violaceous (lilac/purple) discoloration of the upper eyelids, often with periorbital edema, named for the purple heliotrope flower. It is highly specific for dermatomyositis and is one of the recognized cutaneous hallmarks. The rash is photosensitive and frequently accompanies other DM skin findings (Gottron, shawl/V signs). Its presence makes the diagnosis DM rather than PM even if muscle findings are identical.

BOARD TRAP

WRONG answer: labeling it a malar (butterfly) rash and diagnosing SLE. Discriminator: the malar rash of lupus spares the nasolabial folds and centers on the cheeks/bridge of nose; the heliotrope rash is PERIORBITAL on the eyelids and is violaceous, not pink-red — eyelid involvement points to DM, not lupus.

3. Gottron papules

WHAT YOU SEE

A label arrowing to violaceous papules sitting directly on her knuckles (MCP/PIP/DIP) — the rash drawn ON the bony knuckle bumps is the visual signature of Gottron papules.

WHAT IT ENCODES

Gottron papules are flat-topped violaceous papules located directly OVER the bony extensor surfaces — MCP, PIP, and DIP joints (also elbows, knees). Together with the heliotrope rash they are considered pathognomonic for dermatomyositis. The flat macular form over the same joints is called Gottron sign. They reflect the photosensitive, interface dermatitis of DM.

BOARD TRAP

WRONG answer: confusing the distribution with lupus. Discriminator: DM rash sits ON the knuckles (Gottron papules over the MCP/PIP joints) and SPARES the skin between joints; SLE does the reverse — it spares the knuckles and affects the phalanges between the joints. 'Over the joints = DM, between the joints = lupus.'

4. DM patient

WHAT YOU SEE

The woman in the apron stamped 'DM' on the left counter — she is the dermatomyositis figure, carrying the skin findings (heliotrope, Gottron) that the man labelled PM lacks.

WHAT IT ENCODES

Dermatomyositis = characteristic SKIN findings (Gottron, heliotrope, shawl/V signs) PLUS proximal symmetric weakness. Biopsy shows PERIFASCICULAR atrophy with PERIMYSIAL/perivascular CD4+ and B-cell infiltrate and complement (MAC) deposition on capillaries — a humoral, microangiopathic process. Adult DM carries a markedly increased malignancy risk, so every adult needs an age- and risk-appropriate cancer screen at diagnosis.

BOARD TRAP

WRONG answer: making the diagnosis DM but then skipping the cancer hunt because the patient 'looks well.' Discriminator: adult DM is strongly paraneoplastic (ovarian, lung, GI, nasopharyngeal, breast) — failure to perform age-appropriate malignancy screening (and consider CT chest/abdomen/pelvis ± pelvic/mammography) is the classic management error.

5. Juvenile + calcinosis

WHAT YOU SEE

The small child figure at lower left labelled 'JUVENILE + CALCINOSIS' — the kid stands for juvenile DM, and the calcinosis tag marks the subcutaneous calcium deposits that drive the pediatric form (no cancer link).

WHAT IT ENCODES

Juvenile dermatomyositis (JDM) is the pediatric form, characterized by a vasculopathy and a strong tendency toward CALCINOSIS CUTIS (subcutaneous calcium deposits) and lipodystrophy. Unlike adult DM, JDM is NOT associated with underlying malignancy. Anti-NXP2 and anti-MDA5 antibodies are seen in children; treatment is steroids plus steroid-sparing agents to limit calcinosis.

BOARD TRAP

WRONG answer: ordering an exhaustive paraneoplastic/cancer workup in a child with DM. Discriminator: the malignancy association is an ADULT phenomenon — juvenile DM is not paraneoplastic, so calcinosis and vasculopathy, not cancer, drive the workup and prognosis.

6. Always biopsy

WHAT YOU SEE

The central sign reading 'ALWAYS BIOPSY' — the imperative banner makes muscle biopsy the gold-standard confirmatory step that tells the subtypes apart.

WHAT IT ENCODES

Muscle biopsy remains the gold-standard confirmatory test for inflammatory myopathy and distinguishes subtypes by pattern: perifascicular atrophy (DM), endomysial CD8+ invasion of non-necrotic fibers (PM), rimmed vacuoles (IBM), and scattered necrosis with sparse inflammation (necrotizing myopathy). Pick a moderately weak muscle (not end-stage); EMG/MRI can guide the site. Shared marker = sarcolemmal MHC-1 upregulation.

BOARD TRAP

WRONG answer: starting long-term immunosuppression on the basis of an autoantibody panel alone without tissue (or, where appropriate, a confirmatory MRI/EMG). Discriminator: many mimics (steroid myopathy, statin toxicity, hypothyroid myopathy, IBM) require tissue or other confirmation — committing a patient to chronic steroids without confirming inflammatory myopathy is the trap.

7. PM patient

WHAT YOU SEE

The man in the apron stamped 'PM' standing at the central counter — he is the polymyositis figure: proximal weakness but NO skin findings (bare skin, unlike the DM woman).

WHAT IT ENCODES

Polymyositis = proximal symmetric weakness WITHOUT the DM skin findings. Biopsy shows ENDOMYSIAL CD8+ cytotoxic T cells invading non-necrotic, MHC-1-expressing fibers (a cell-mediated process, contrasting with DM's humoral perimysial CD4 pattern). PM is a diagnosis of exclusion — it is over-diagnosed, and many 'PM' cases are actually IBM, necrotizing myopathy, or dystrophy.

BOARD TRAP

WRONG answer: diagnosing PM in a patient who actually has subtle cutaneous disease. Discriminator: if ANY characteristic skin finding is present (even subclinical Gottron or nailfold changes), the diagnosis is dermatomyositis, not polymyositis — examine the skin and nailfolds carefully before settling on PM.

8. If normal → think IBM

WHAT YOU SEE

The arrow/sign near the central CK gauge reading 'if NORMAL → think IBM' — a normal/near-normal CK should redirect you off DM/PM toward inclusion body myositis.

WHAT IT ENCODES

A CK that is only mildly elevated or near-normal, combined with a slowly progressive, treatment-RESISTANT course, should redirect you toward inclusion body myositis. IBM classically affects men over 50, with distal (finger flexor) and asymmetric (quadriceps) weakness, frequent dysphagia, and rimmed vacuoles with congophilic/p62/TDP-43 inclusions on biopsy.

BOARD TRAP

WRONG answer: continuing to escalate steroids/immunosuppression in a 'PM' patient who is not responding. Discriminator: lack of response plus a near-normal CK and distal/asymmetric weakness = IBM, which is essentially refractory to immunosuppression — more steroids only add steroid myopathy and side effects.

9. CK <10x

WHAT YOU SEE

The CK gauge/meter beside the central PM figure pointing to a moderate '<10x' zone — DM/PM raise CK but typically under 10x, unlike the sky-high necrotizing range.

WHAT IT ENCODES

Creatine kinase (CK) is the most useful serum marker of muscle injury and is typically markedly elevated — often more than 10x the upper limit of normal — in active DM/PM, paralleling disease activity. Other muscle enzymes (aldolase, AST/ALT, LDH) also rise. CK is used to track flares but treatment is titrated to STRENGTH, not the number.

BOARD TRAP

WRONG answer: excluding myositis because the CK is normal, or treating purely off the CK. Discriminator: a normal CK does NOT rule out inflammatory myopathy (DM can have normal CK, and IBM often does) — rely on the clinical exam, EMG, MRI, and biopsy rather than the enzyme alone.

10. Anti-Mi-2

WHAT YOU SEE

The 'Mi-2' antibody badge in the bottom row — the marker tagged with a favorable / good-prognosis cue, flagging the classic skin-predominant DM with LOW cancer risk.

WHAT IT ENCODES

Anti-Mi-2 is a myositis-SPECIFIC antibody that defines a classic dermatomyositis phenotype with prominent cutaneous features (Gottron, heliotrope, V/shawl signs), a GOOD response to immunosuppression, and a comparatively LOW malignancy risk. It is one of the more favorable serologic groups in DM.

BOARD TRAP

WRONG answer: equating all DM antibodies with high cancer risk and ordering aggressive surveillance for an anti-Mi-2 patient. Discriminator: anti-Mi-2 carries low malignancy risk and good prognosis — the high-cancer antibodies are anti-TIF1-gamma and anti-NXP2, not anti-Mi-2.

11. Anti-TIF1-gamma

WHAT YOU SEE

The 'TIF1-γ' antibody badge along the bottom row — the marker carrying the strongest cancer-screening cue, flagging the highest malignancy risk in adult DM.

WHAT IT ENCODES

Anti-TIF1-gamma (anti-p155/140) and anti-NXP2 are the dermatomyositis antibodies most strongly linked to underlying MALIGNANCY in adults — TIF1-gamma positivity warrants thorough, often repeated, cancer screening. TIF1-gamma is also associated with severe cutaneous disease. (Anti-MDA5, by contrast, marks clinically amyopathic DM with rapidly progressive ILD and skin ulcers.)

BOARD TRAP

WRONG answer: reassuring an anti-TIF1-gamma-positive adult that cancer is unlikely. Discriminator: this antibody carries the HIGHEST malignancy association in adult DM — it should trigger comprehensive, age-appropriate cancer screening (CT chest/abdomen/pelvis, mammography, pelvic exam, colonoscopy as indicated), not reassurance.

12. CK severity dial

WHAT YOU SEE

The horizontal gauge at the bottom sweeping from 'CK normal' (IBM end) up through DM/PM to 'IMNM >10x' — the dial ranks subtypes by how high the CK climbs.

WHAT IT ENCODES

CK magnitude roughly maps to subtype: NORMAL or near-normal in IBM (and amyopathic DM), MILDLY-to-moderately elevated in DM, and VERY HIGH (>10x ULN) in immune-mediated necrotizing myopathy (IMNM, anti-SRP/anti-HMGCR). Polymyositis sits in the high range as well. The dial is a quick way to rank suspicion by enzyme level.

BOARD TRAP

WRONG answer: reflexively calling a sky-high CK 'polymyositis.' Discriminator: the highest CK levels (often >10–50x) belong to necrotizing myopathy (anti-SRP/anti-HMGCR), while a low/normal CK with weakness points to IBM — let the CK magnitude steer subtype, not just the label PM.

13. 5-yr survival 70-85%

WHAT YOU SEE

The sign reading '5-yr survival 70-85%' near the juvenile corner — the number reminds you the disease has real mortality, driven by lung/cardiac/cancer complications, not the limb weakness.

WHAT IT ENCODES

Inflammatory myopathy carries meaningful mortality (roughly 70–85% 5-year survival in older series), driven not by limb weakness itself but by EXTRAMUSCULAR complications: interstitial lung disease, cardiac involvement (myocarditis, arrhythmia), aspiration from pharyngeal weakness, and associated MALIGNANCY. Prognosis improves with early treatment and cancer detection.

BOARD TRAP

WRONG answer: assuming inflammatory myopathy is a benign 'just weakness' disease. Discriminator: prognosis is dictated by ILD, cardiac disease, dysphagia/aspiration, and malignancy — overlooking these systemic associations, not the muscle weakness, is what kills patients.

14. Shawl/V-sign rash

WHAT YOU SEE

The red rash drawn as a 'V' over the upper chest/neck and a shawl pattern across the shoulders on the torso — these photodistributed sun-exposed rashes are the V-sign and shawl-sign of DM.

WHAT IT ENCODES

The V-sign (photodistributed erythema over the anterior neck/upper chest) and the shawl sign (erythema across the shoulders/upper back) are photosensitive cutaneous signs of dermatomyositis. Like the heliotrope and Gottron findings, they are sun-exposed and reinforce a DM (not PM) diagnosis; the holster sign (lateral thighs) is the analogous lower-body finding.

BOARD TRAP

WRONG answer: dismissing a photodistributed chest/back rash as ordinary sun damage or polymorphous light eruption. Discriminator: in a patient with proximal weakness, a violaceous V-sign or shawl-sign distribution is a dermatomyositis skin marker and changes the diagnosis from PM to DM (and triggers the malignancy screen).
